WCCM-ECCOMAS 2026 | MS165 - METHODS AND APPLICATIONS OF MODEL ORDER REDUCTION

Hybrid Green's Function Learning With Axial Reduction

A two-dimensional elliptic solver built from axial Green operators and a learned directional source split

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REDUCE
a multi-dimensional elliptic problem into axial Green subproblems

COUPLE
learn the directional source split that makes axial reconstructions consistent

RECOVER
a multi-dimensional solution from coupled axial Green operators

Axial Reduction of Elliptic Operators

Replace a global source-to-solution map with structured one-dimensional Green reconstructions

$$\begin{aligned} -\nabla\cdot(a\nabla u) + \mathbf{b}\cdot\nabla u + cu &= f \quad \text{in } \Omega, \\ u &= 0 \quad \text{on } \partial\Omega, \end{aligned} \qquad \mathbf{b} = (b_x, b_y).$$

$$\mathcal{S}_{a,\mathbf{b},c} : f \mapsto u, \qquad u = \mathcal{S}_{a,\mathbf{b},c}[f].$$

Operator fixed; source varies.

DIRECT MAP
one global source-to-solution representation
High-dimensional and geometry-sensitive

→

REDUCED VIEW
directional 1-D operators + learned coupling
Structure-preserving MOR framing

$$L_x u = -\partial_x(a\partial_x u) + b_x\partial_x u + \frac{1}{2}cu, \qquad L_y u = -\partial_y(a\partial_y u) + b_y\partial_y u + \frac{1}{2}cu.$$

$$L_x u + L_y u = f.$$

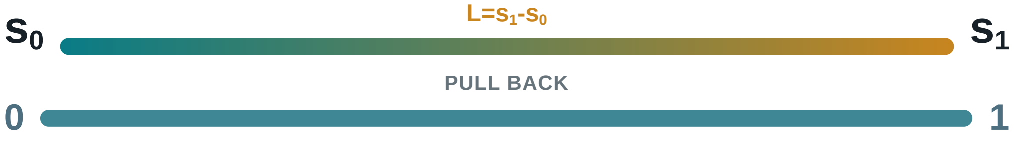
Unit-Interval Pull-Back and Operator Scaling

Physical length is absorbed into normalized operator coefficient profiles and source profile

PHYSICAL ONE-DIMENSIONAL AXIAL OPERATOR

$$s \in [s_0, s_1],$$
$$\mathcal{L}_{\text{phys}} u = -\frac{d}{ds}\left(a_{\text{phys}}(s)\frac{du}{ds}\right) + b_{\parallel,\text{phys}}(s)\frac{du}{ds} + c_{\text{phys}}(s)u = f_{\text{phys}}(s).$$

NORMALIZE THE COORDINATE



$$s = s_0 + Lt, \qquad v(t) = u(s_0 + Lt), \qquad t \in [0,1].$$

The interval is normalized, but the physical length is not discarded.

$$a_{\text{unit}} = a_{\text{phys}}, \quad a'_{\text{unit}} = La'_{\text{phys}}, \quad b_{\parallel,\text{unit}} = Lb_{\parallel,\text{phys}}, \quad c_{\text{unit}} = L^2 c_{\text{phys}}, \quad f_{\text{unit}} = L^2 f_{\text{phys}}.$$

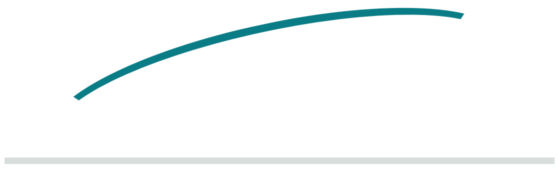
$$-\frac{d}{dt}\left(a_{\text{unit}}(t)\frac{dv}{dt}\right) + b_{\parallel,\text{unit}}(t)\frac{dv}{dt} + c_{\text{unit}}(t)v(t) = f_{\text{unit}}(t), \qquad t \in [0,1].$$

Here $b_\parallel = b_x$ on an x-directed interval and $b_\parallel = b_y$ on a y-directed interval.

GreenNet I: Normalized Axial Green Operator

Learn one-dimensional Green kernels, not a global two-dimensional kernel

SOURCE PROFILE

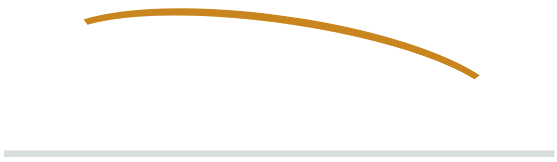


$f_{\text{unit}}(\eta)$

$G_{\text{unit}}(t, \eta)$

KERNEL INTEGRAL OPERATOR

SOLUTION PROFILE



$v(t)$

$$\mathcal{G}_{\text{unit}}[f_{\text{unit}}](t) = \int_0^1 G_{\text{unit}}(t, \eta) f_{\text{unit}}(\eta)\, d\eta, \qquad v = \mathcal{G}_{\text{unit}}[f_{\text{unit}}].$$

Operator coefficients: Axial coefficient profiles define each local 1D operator.

Kernel coordinates (t, η) define the evaluation and source points.

GreenNet target is a local unit-interval kernel, not a global 2D Green function.

GreenNet II: Analytic Green Structure and Learned Correction

A hybrid kernel: analytic singular structure plus neural smooth correction

HYBRID GREEN KERNEL

$$G_\theta = \text{analytic jump} + \text{analytic cancellation} + \text{learned smooth correction.}$$

ANALYTIC JUMP

Build the Dirac-() structure.

ANALYTIC CANCELLATION

Cancel Heaviside-type contributions.

LEARNED CORRECTION

Represent the remaining smooth residual.

GreenNet II: Analytic Green Structure and Learned Correction

A hybrid kernel: analytic singular structure plus neural smooth correction

$$G_\theta(t,\eta) = \underbrace{E(t,\eta)M(t)R_\theta(t,\eta)}_{\text{learned smooth correction}} + \underbrace{B(t)\left(J_0(t,\eta) - \frac{1}{2}E(t,\eta)\right)}_{\text{Heaviside cancellation}} + \underbrace{A(t)G_0(t,\eta)}_{\text{Dirac jump}}.$$

DIRAC JUMP

$$G_0(t,\eta) = \begin{cases} t(1-\eta), & t < \eta, \\ \eta(1-t), & t \geq \eta, \end{cases} \qquad \partial_t^2 G_0 = -\delta(t-\eta).$$

HEAVISIDE CANCELLATION

$$\partial_t J_0(t,\eta) = G_0(t,\eta), \quad S = J_0 - \frac{1}{2}E, \quad \partial_t^2 S = \partial_t G_0.$$

COEFFICIENT FACTORS

$$A(t) = \frac{1}{a_{\text{unit}}(t)}, \qquad B(t) = \frac{a'_{\text{unit}}(t) + b_{\|,\text{unit}}(t)}{a_{\text{unit}}(t)^2}.$$

GreenNet II: Analytic Green Structure and Learned Correction

A hybrid kernel: analytic singular structure plus neural smooth correction

$$G_\theta(t,\eta) = \underbrace{E(t,\eta)M(t)R_\theta(t,\eta)}_{\text{learned smooth correction}} + \underbrace{B(t)\left(J_0(t,\eta) - \frac{1}{2}E(t,\eta)\right)}_{\text{Heaviside cancellation}} + \underbrace{A(t)G_0(t,\eta)}_{\text{Dirac jump}}.$$

LEARNED SMOOTH CORRECTION

$$E(t,\eta)M(t)R_\theta(t,\eta), \qquad E(t,\eta) = t\eta(1-\eta), \qquad M(t) = 1-t.$$

GreenNet learns the smooth correction R_θ; the singular and cancellation structure is built into the kernel form.

GreenNet III: Source-to-Solution Supervision

Supervise the Green operator action, not pointwise exact-kernel labels

$w(t) \sim \mathcal{GP}(0, k_\ell)$ → $v(t) = w(t) - ((1-t)w(0) + t\, w(1))$ so $v(0) = v(1) = 0$ → $f_{\text{unit}} = \mathcal{L}_{\text{unit}} v$

$$v_\theta(t) = \int_0^1 G_\theta(t,\eta) f_{\text{unit}}(\eta)\, d\eta.$$

$$\mathcal{J}_{\text{Green}} \sim \mathbb{E}\left[\int_0^1 |v_\theta(t) - v(t)|^2 \, dt\right].$$

Training checks whether the learned Green kernel maps manufactured sources to target solutions.

CouplingNet I: Source-Conditioned Directional Split

What source should each axial inverse receive?

$G_x[\cdot]$

full source f

?

$G_y[\cdot]$

$$f \longmapsto (\phi, \psi), \qquad \phi + \psi = f.$$

source-conditioned split → x-direction source ϕ | y-direction source ψ

CouplingNet supplies the multi-dimensional coupling that GreenNet alone does not provide.

CouplingNet II: Branches and Local Context for Split Prediction

Source, fixed operator context, axial location, and transverse boundary context

SOURCE BRANCH

forcing profile that changes sample to sample

$$(\mathrm{source}, \mathrm{operator}, t_{\parallel}, t_{\perp})$$

$$\downarrow$$

$$(\phi, \psi)$$

FIXED OPERATOR CONTEXT

diffusion, primary/transverse convection, reaction context

AXIAL LOCAL TRUNK

primary interval coordinate $t_{\parallel}$

POINTWISE TRANSVERSE TRUNK

transverse boundary context $t_{\perp}$

Coefficient profiles are fixed operator context, not sample-varying coefficient inputs.

CouplingNet III: Projection and Green Reconstruction

Physical projection enforces balance before Green reconstruction

RAW SPLIT

$\phi_{\mathrm{raw}}, \psi_{\mathrm{raw}}$

→

PHYSICAL BALANCE PROJECTION

$$r = f - \phi_{\mathrm{raw}} - \psi_{\mathrm{raw}}$$

$$\phi = \phi_{\mathrm{raw}} + \frac{1}{2}r, \quad \psi = \psi_{\mathrm{raw}} + \frac{1}{2}r$$

→

GREEN RECONSTRUCTION

$$u_\phi = G_x[\phi], \qquad u_\psi = G_y[\psi]$$

$$\phi + \psi = f, \qquad u_{\mathrm{pred}} = \frac{1}{2}(u_\phi + u_\psi).$$

Energy-Norm Error Bound Proposition

The split-energy loss is an error-bound quantity under structural assumptions

u_ϕ

$u_{\text{pred}} = \frac{1}{2}(u_\phi + u_\psi)$

u_ψ

$$\|v\|_a^2 = \int_\Omega a|\nabla v|^2, \qquad \mathcal{E}_{\text{split}} = \|u_\phi - u_\psi\|_a^2.$$

$$\|u_{\text{pred}} - u_*\|_a \le \frac{C_E}{2}\sqrt{\mathcal{E}_{\text{split}}}.$$

Conditional structural proposition: exact or controlled Green reconstruction, source-linear weak inverse behavior, and $H_0^1(\Omega)$ admissibility.

Numerical Evidence I: GreenNet Reconstruction Quality

The normalized Green operator reconstructs interval solutions from sources

FIGURE SLOT: GREENNET INTERVAL RECONSTRUCTION

source profile

target vs reconstruction

signed error

$$v_\theta(t) = \int_0^1 G_\theta(t,\eta) f_{\mathrm{unit}}(\eta)\, d\eta, \qquad v_\theta(t) \approx v(t).$$

Final figure and performance wording will be selected after GreenNet result plots are available.

Numerical Evidence II: CouplingNet Solution Reconstruction

The learned split and Green reconstruction produce a multi-dimensional solution

FIGURE SLOT: COUPLINGNET FULL SOLUTION RECONSTRUCTION

reference	prediction	signed error	split mismatch

$$u_{\text{pred}} = \frac{1}{2}(u_\phi + u_\psi), \qquad u_{\text{pred}} - u_*, \qquad u_\phi - u_\psi.$$

Keep the split diagnostic only if the final figure remains readable.

Contributions and Takeaway

A structured reduced operator learner for elliptic PDEs

Learn axial Green inverses, learn the coupling, and recover a consistent multidimensional solution.

REDUCE

multi-dimensional elliptic learning to axial Green operators

EMBED

analytic Green singular/cancellation structure

COUPLE

source-conditioned split using source/operator/local context

BOUND

split-energy consistency connects to final solution error

The key reduction is not only geometric; it is operator-structural.

Backup / Q&A Menu

Details available if the audience asks

READY WITHOUT FIGURES

- Backup 14A: Dirac/Heaviside derivation sketch
- Backup 14B: imperfect Green reconstruction perturbation
- Backup 14C: connected-interval pull-back detail

DEFERRED UNTIL FIGURES ARE SELECTED

- extra GreenNet reconstruction evidence
- extra CouplingNet split/energy evidence

Backup 14A: Dirac/Heaviside Derivation Sketch

Why the analytic Green wrapping has three terms

$$G_\theta = EMR_\theta + B\left(J_0 - \frac{1}{2}E\right) + AG_0.$$

DIRAC JUMP

$$\partial_t^2 G_0(t,\eta) = -\delta(t-\eta)$$

HEAVISIDE CANCELLATION

$$\partial_t J_0(t,\eta) = G_0(t,\eta)$$

$$S = J_0 - \frac{1}{2}E, \qquad \partial_t^2 S = \partial_t G_0$$

SMOOTH RESIDUAL

The neural term learns what remains after analytic singular and cancellation structure is built in.

Backup 14B: Imperfect Green Reconstruction Perturbation

What changes when the learned Green inverse is not exact?

$$G_x[\phi_*] = u_* + \varepsilon_x, \qquad G_y[\psi_*] = u_* + \varepsilon_y.$$

$$\|u_{\mathrm{pred}} - u_*\|_a \le \frac{C_E}{2}\left(\sqrt{\mathcal{E}_{\mathrm{split}}} + \|\varepsilon_x - \varepsilon_y\|_a\right) + \frac{1}{2}\|\varepsilon_x + \varepsilon_y\|_a.$$

$\varepsilon_x - \varepsilon_y$ perturbs split consistency.

$\varepsilon_x + \varepsilon_y$ is common Green bias and can remain invisible to split consistency.

Backup 14C: Connected-Interval Pull-Back Detail

How non-square domains become normalized one-dimensional Green problems

CONNECTED PHYSICAL INTERVAL

$$I = [s_0, s_1], \qquad s = s_0 + Lt, \qquad t \in [0,1].$$

$$a_{\mathrm{unit}} = a_{\mathrm{phys}}, \quad a'_{\mathrm{unit}} = La'_{\mathrm{phys}}, \quad b_{\parallel,\mathrm{unit}} = Lb_{\parallel,\mathrm{phys}}, \quad c_{\mathrm{unit}} = L^2 c_{\mathrm{phys}}, \quad f_{\mathrm{unit}} = L^2 f_{\mathrm{phys}}.$$

Each connected interval is an independent one-dimensional Dirichlet Green problem.
Disconnected intervals are not merged, because that would create artificial information flow outside the domain.